Type MS · Sum to infinity of a geometric series

Answer 1 of 13 · Question 8 part c · 2 marks

2024 JUN · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

8 (a)	$a+ar=400$ or $\frac{a(1-r^2)}{1-r}=400,\quad ar+ar^2=100$ oe	B1B1
	Common methods 1) $r(a+ar)=100 \Rightarrow r(400)=100$	M1
	$r=\frac{1}{4}$*	A1cso
	2) $a=\frac{400}{1+r} \Rightarrow \left(\frac{400}{1+r}\right)r+\left(\frac{400}{1+r}\right)r^2=100$	M1
	$\Rightarrow 400r+400r^2=100+100r \Rightarrow 4r^2+3r-1=0 \Rightarrow (4r-1)(r+1)(=0)$ $r=\frac{1}{4}$*	A1cso
	3) $r=\frac{400-a}{a} \Rightarrow a\left(\frac{400-a}{a}\right)+a\left(\frac{400-a}{a}\right)^2=100 \Rightarrow$	M1
	$\Rightarrow 400-a+\frac{(400-a)^2}{a}=100 \Rightarrow$ $400-a^2+160000-800a+a^2=100a \Rightarrow a=320 \Rightarrow 320+320r=400$ $r=\frac{1}{4}$*	A1cso [4]
ALT	(Let G_1,G_2,G_3 be the first 3 terms) $G_2+G_3=100$	B1
	$(G_1+G_2=400) \Rightarrow rG_1+rG_2=100$	B1
	$rG_1+rG_2=100 \Rightarrow (r(G_1+G_2)=100) \Rightarrow r(400)=100$	M1
	$r=\frac{1}{4}$ *	A1*cso [4]
(b)	$(a=)\frac{300}{1-\left(\frac{1}{4}\right)^2}$ or $\frac{400}{1+\frac{1}{4}}$ or $\frac{100}{\frac{1}{4}+\left(\frac{1}{4}\right)^2}$ $=320$ *	M1A1 cso [2]
(c)	$S_\infty=\frac{320}{1-\frac{1}{4}}=\frac{1280}{3}$	M1A1 [2]
		Total 12 marks

Type MS · Sum to infinity of a geometric series

Answer 1 of 13 · Question 8 part c · 2 marks

2024 JUN · Paper 01 · extract 2 of 4 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	B1	For either equation shown correct a and r can be any letters throughout.
	B1	For both equations shown correct a and r can be any letters throughout.
	M1	For forming an equation eliminating a or r Allow one error in processing **such as a sign or arithmetical error, but not a 'cancellation'/simplification error**. Must be working with 2 correct equations. This mark can be awarded as soon as a or r are eliminated. Doesn't need simplification at this stage.
	A1 cso	For correctly solving and attaining $r = \frac{1}{4}$ minimum steps shown, no errors/omissions, ignore $r = -1$
ALT	B1	For either equation shown correct
	B1	For both equations shown correct
	M1	For multiplication of the first equation by r and formation of an equation in r Allow one error in processing. Must be working with 2 correct equations.
	A1	For $r = \frac{1}{4}$ minimum steps shown, no errors or omissions. Ignore work on any negative values
There are a number of different methods to do this, the four most commonly anticipated are shown. Mark to the following principles to gain the method mark: • One processing error only in any method (**M mark only, not A mark).** • Rearrange for r or a and correctly substitute into the other equation or such as method 1 to reach an equation in one variable only. • Rearrange the resulting equation so that an equation of the form $\boldsymbol{br = c}$ is reached. Note for the quadratic option, a factorisation will suffice. Note, if eliminating r, a must be found and the value of a then substituted into an appropriate equation. Methods where these principles can't be applied and thought worthy of credit – **send to review please**.		
(b)	M1	For using their expression for a with the correct r, to find a value for a Note, **for this question only**, work in (a) may be credited for this mark – **only if they eliminated r in their solution for part (a) and this is then used.**
	A1cso	For 320, no errors.
(c)	M1	For using the correct formula for the sum to infinity of a convergent series with the given values of a and r to find a value.
	A1	For the exact value of $\frac{1280}{3}$ oe or 426.67 or better (ie correctly rounded to more decimal places) or 426.6… (minimum 3 dots) or 426.66^{r} or $426.\dot{6}$
(d)	M1	Uses the correct formula for the sum of a geometric series, to set up **an inequality or equation,** allow < or > or = using the given values of r and a. Condone $\frac{1}{4}^{n}$
	dM1	For simplifying (allow errors in simplification) their **inequality or equation** in n to the form $\left(\frac{1}{4}\right)^{n} < d \quad d \neq 0$ or $4^{n} < d$ Allow < or > or =. Dependent on the 1st method mark. Condone poor bracketing with powers again.
	ddM1	For the correct use of logs and correct use of an inequality sign throughout, including the reversal at the appropriate point. Dependent on both previous method marks. This mark may not be awarded if 'd' is negative. If candidates give a final answer of (n =) 7 – this mark can be implied even if the inequality sign is not correctly reversed.
	A1	For (n =) 7 Note although $n = 7$ can imply ddM1 as described, it is unlikely to imply the first 2 marks as there must be some logs work (directed by the question).

Type MS · Sum to infinity of a geometric series

Answer 1 of 13 · Question 8 part c · 2 marks

2024 JUN · Paper 01 · extract 3 of 4 · find part c in this mark scheme extract

(d)	$426.6 < \dfrac{320\left(1-\left(\frac{1}{4}\right)^n\right)}{1-\frac{1}{4}} \Rightarrow \left(\frac{1}{4}\right)^n < \frac{1}{6400}$ or $4^n < 6400$	M1dM1
	$\Rightarrow n > \dfrac{\log\left(\frac{1}{6400}\right)}{\log\left(\frac{1}{4}\right)}$ or $n > \log_{\frac{1}{4}}\left(\frac{1}{6400}\right)$ oe	ddM1
	$\Rightarrow n > 6.32... \Rightarrow n = 7$	A1 [4]
		Total 12 marks

Type MS · Sum to infinity of a geometric series

Answer 1 of 13 · Question 8 part c · 2 marks

2024 JUN · Paper 01 · extract 4 of 4 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Sum to infinity of a geometric series

Answer 2 of 13 · Question 3 part b · 2 marks

2022 JAN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

3(a)(i)	$\frac{ar^4}{ar} = \frac{\frac{135}{1024}}{\frac{5}{16}}$ $r = \sqrt[3]{\frac{\frac{135}{1024}}{\frac{5}{16}}} = \left(\sqrt[3]{\frac{135\times16}{5\times1024}}\right) = \ldots$ $r = \frac{3}{4}$ oe	M1 M1 A1
(a)(ii)	$ar = \frac{5}{16} \Rightarrow a = \frac{\frac{5}{16}}{\frac{3}{4}} = \left(\frac{5}{12}\right)$ $a = \frac{5}{12}$	M1 A1 [5]
(b)	$S = \frac{\frac{5}{12}}{1-\frac{3}{4}} = \ldots$ $S = \frac{5}{3}$	M1 A1 [2]
		Total 7 marks

Type MS · Sum to infinity of a geometric series

Answer 2 of 13 · Question 3 part b · 2 marks

2022 JAN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

(b)	**M1**	For using the correct formula for the sum to infinity using their a and r provided $\|r\| < 1$ $S = \frac{'\frac{5}{12}'}{1-'\frac{3}{4}'} = \ldots$ $\frac{'5'}{12} \neq \frac{5}{16}$, $\frac{'5'}{12} \neq \frac{135}{1024}$
	A1	For the correct value of $S = \frac{5}{3}$ **Note:** Must be the exact value.

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Sum to infinity of a geometric series

Answer 2 of 13 · Question 3 part b · 2 marks

2022 JAN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Sum to infinity of a geometric series

Answer 3 of 13 · Question 5 part a · 2 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

Number		
5 (a)	$[S_\infty =]\dfrac{12}{1-\frac{3}{8}} = \dfrac{96}{5}$	M1 A1 (2)
(b)	$ar^5 = 12\left(\dfrac{3}{8}\right)^5$	M1
	$= \dfrac{2^2\times 3\times 3^5}{2^{15}} = \dfrac{3^6}{2^{13}}$ *	M1 A1cso (3)
(c)	e.g. $u_n = 12\left(\dfrac{3}{8}\right)^{n-1}$	M1
	$\log_2 u_n = \log_2 12 + \log_2\left(\dfrac{3}{8}\right)^{n-1}$	M1
	$\log_2 u_n = \log_2 12 + (n-1)\left[\log_2 3 - \log_2 8\right]$	M1
	$\log_2 u_n = \log_2 3 + 2\log_2 2 + (n-1)[\log_2 3 - 3\log_2 2]$	M1
	$\log_2 u_n = n\log_2 3 - 3n + 5$ *	A1 cso (5)
		Total 10 marks

Type MS · Sum to infinity of a geometric series

Answer 3 of 13 · Question 5 part a · 2 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For the correct use of $\frac{a}{1-r}$
	A1	For the correct value $\frac{96}{5}$
(b)	M1	For the correct use of ar^{n-1}
	M1	For expressing 12 $\left(2^2\times 3\right)$ and 8 $\left(2^3\right)$ as powers of 2
	A1cso	For obtaining the given expression with no errors seen.
(c)	M1	For the correct use of ar^{n-1} and the given values of a and r to give $u_n = 12\left(\frac{3}{8}\right)^{n-1}$
	M1	For taking logs [base 2] of both sides and applying the addition law.
	M1	For applying the power law and subtraction laws to $\log_2\left(\frac{3}{8}\right)^{n-1}$ $\log_2\left(\frac{3}{8}\right)^{n-1} = (n-1)\left[\log_2 3 - \log_2 8\right]$
	M1	For obtaining $\log_2 12 = \log_2 3 + 2\log_2 2$
	A1cso	For obtaining the given equation with no errors seen.
	ALT	
	M1	For the correct use of ar^{n-1} and the given values of a and r to give $u_n = 12\left(\frac{3}{8}\right)^{n-1}$
	M1	For rearranging the equation to obtain: $U_n = 12\left(\frac{3}{8}\right)^{n-1} = 2^2\times 3\times\frac{3^{n-1}}{2^{3(n-1)}} = \frac{3^n}{2^{3n-5}}$
	M1	For taking logs of both sides and applying the addition (subtraction) law. $\log_2 U_n = \log_2 3^n - \log_2\left(3n-5\right)$
	M1	For applying the power law to obtain: $\log_2 U_n = n\log_2 3 - \log_2\left(3n-5\right)$
	A1 cso	For obtaining the given equation with no errors seen.

Type MS · Sum to infinity of a geometric series

Answer 3 of 13 · Question 5 part a · 2 marks

2022 JAN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sum to infinity of a geometric series

Answer 4 of 13 · Question 7 part c · 2 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

7(a)	$\frac{(x+1)}{(x-3)}=\frac{(4x-2)}{(x+1)}$ or $(x+1)^2=(x-3)(4x-2)$	M1
	$(x+1)^2=(4x-2)(x-3)\Rightarrow 3x^2-16x+5(=0)$	A1
	$(x-5)(3x-1)=0\Rightarrow x=5,\ \frac{1}{3}$	M1A1A1 (5)
(b)	$x=\frac{1}{3}\Rightarrow r=\frac{x+1}{x-3}=\frac{\frac{4}{3}}{\frac{-8}{3}}$	M1
	$r=-\frac{1}{2}$ ∴ convergent as $-1<r<1$ *	A1cso (2)
(c)	$S=\frac{a}{1-r}=\frac{-\frac{8}{3}}{1+\frac{1}{2}}=-\frac{16}{9}$ oe	M1A1 (2)
(d)	$\frac{S}{S_n}=\frac{a}{1-r}\times\frac{1-r}{a(1-r^n)}=\frac{1}{(1-r^n)}=\frac{256}{255}$	M1
	$255=256(1-r^n)$	
	$256r^n=1$ $\left(-\frac{1}{2}\right)^n=\frac{1}{256}$ oe $n=8$	dM1A1 (3) [12]

(a)M1	Form an equation using the given information about the terms
A1	Simplify their equation to a correct 3TQ, terms in any order Condone missing = 0
M1	Attempt to solve their 3TQ by any valid method. Must reach x = ... (at least one root) Calculator solutions: *Both* roots correct from a correct equation scores M1A1A1 Incorrect equation or incorrect roots scores M0A0A0
A1	One correct value of x
A1	Both correct values of x
(b)	
M1	Use either of their values of x, provided it is < 1, to find the corresponding value of r. No need to simplify
A1cso	Correct value of r **and** the conclusion including the reason
(c)	
M1	Use their value of r (not x) provided $-1<r<1$ (as found in (b) or here) and the formula for the sum to infinity to obtain a value for S
A1	Correct value
(d)	
M1	Obtain an equation in r and n. May use the formulae to cancel a or may sub values of a and r in the formulae for the LHS Must equate to 256/255 Value of r not needed for this mark so allow any value used.
dM1	Solve their equation of the form r^n = ... where $-1<r<1$ (r not x). May use trial and improvement or logs. This mark can be given if the equation and value of r are incorrect. Evidence of method needed if final answer is incorrect. If logs used condone log $(-1/2)^n$
A1	Correct value from correct working

Type MS · Sum to infinity of a geometric series

Answer 4 of 13 · Question 7 part c · 2 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question Number	Scheme	Marks
	5	

	M1	$\left(6\sqrt{2}\right)$ or AC $\left(12\sqrt{2}\right)$
	M1	For using the correct trigonometry (tan ratio): $h = '6\sqrt{2}'\tan 30^\circ = \left[2\sqrt{6}\right]$ or equivalent. Eg $\tan 30 = \dfrac{h}{6\sqrt{2}} \Rightarrow h = ...$
	A1	For $2\sqrt{6}$ oe
(b)	M1	For correct expression for the midpoint of AD to F $\left(6 - \dfrac{12}{1+4}\right)$ or 3.6 oe seen
	M1	For the correct use of Pythagoras' to find OF $\dfrac{6\sqrt{34}}{5}$ oe [6.997..] ft their 3.6
	ALT	
	M1	Finds the length OF using cosine rule. $OF = \sqrt{2.4^2 + \left(6\sqrt{2}\right)^2 - 2\times 2.4\times 6\sqrt{2}\times\cos 45} = \left[\dfrac{6\sqrt{34}}{5}\right]$ OR $OF = \sqrt{9.6^2 + \left(6\sqrt{2}\right)^2 - 2\times 9.6\times 6\sqrt{2}\times\cos 45} = \left[\dfrac{6\sqrt{34}}{5}\right]$ OR $\cos 45^\circ = \dfrac{9.6^2 + \left(6\sqrt{2}\right)^2 - OF^2}{2\times 9.6\times 6\sqrt{2}} \Rightarrow OF = ...$ oe [6.997..]
	M1	For the correct evaluation of their cosine rule. $\left(\dfrac{6\sqrt{34}}{5}\text{ oe}\right)$ Accept awrt 7.00 [6.997142…]
	dM1	For the correct use of $\tan\theta = \dfrac{'EO'}{'OF'}$ This mark is dependent on both previous M marks. They must have a valid method to find OF for the award of this mark.
	A1	For 35(°) or better (Calculator value is 34.997…°.)
	Note: There are other methods – if unsure, send to Review.	

Type MS · Sum to infinity of a geometric series

Answer 5 of 13 · Question 6 part b · 3 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks
6 a	$r=\frac{q(2p-3)}{q(2p+3)}=\frac{q(2p+3)}{q(4p+1)}$	M1
	$(4p+1)(2p-3)=(2p+3)^2$	dM1
	$2p^2-11p-6=0$	A1
	$(2p+1)(p-6)=0$	M1
	$p=-\frac{1}{2}$ or $p=6$	A1 (5)
b	When $p=6$ $r=\frac{3}{5}$ and $U_1=q\left(4\times'6'+1\right)=\left(25q\right)$	M1
	$S_\infty=\frac{25q}{\frac{2}{5}}=250\Rightarrow q=4$	dM1 A1 (3)
		Total 8 marks

Type MS · Sum to infinity of a geometric series

Answer 6 of 13 · Question 8 part c · 2 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

8(a)	$a + ar = 360$ oe e.g. $\frac{a(1-r^2)}{1-r} = 360$ $ar + ar^2 = 288$ oe $r = \left\{\frac{a(1+r)}{ar(1+r)} =\right\} \frac{288}{360} = \frac{4}{5}$ $a = \frac{360}{1 + "\frac{4}{5}"} = 200$ $U_n = 200\left(\frac{4}{5}\right)^{n-1}$ *	M1M1 (B1B1on ePen dM1A1 M1A1 A1cso(B1 on ePen) [7]
8(a) ALT	$a + ar = 360$ oe e.g. $\frac{a(1-r^2)}{1-r} = 360$ $ar + ar^2 = 288$ oe $a\left(\frac{360}{a} - 1\right) + a\left(\frac{360}{a} - 1\right)^2 = 288$ $\Rightarrow a = 200$ $r = \frac{360}{"200"} - 1 = \frac{4}{5}$ $U_n = 200\left(\frac{4}{5}\right)^{n-1}$ *	M1M1 (B1B1 on ePen) dM1A1 M1A1 A1cso(B1 on ePen) [7]
(b)	{S is convergent because} $\left\|\frac{4}{5}\right\| < 1$	B1 [1]
(c)	$S_\infty = \frac{'200'}{1-'\left(\frac{4}{5}\right)'} = 1000$	M1A1cao [2]
(d)	$\frac{200\left(1-\left(\frac{4}{5}\right)^n\right)}{1-\left(\frac{4}{5}\right)} > 978$ $\left(\frac{4}{5}\right)^n < \frac{11}{500}$ $n\,log\left(\frac{4}{5}\right) < log\left(\frac{11}{500}\right)$ $\Rightarrow n > 17.104\ldots$ $n = 18$	M1 dM1 ddM1 A1 [4]
		Total 14 marks

Type MS · Sum to infinity of a geometric series

Answer 6 of 13 · Question 8 part c · 2 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	M1	For either correct equation. Allow use of U_1 as the first term instead of a
	M1	For both correct equations. Allow use of U_1 as the first term instead of a
	dM1	For eliminating a from their equations, reaches r =... (Depends on previous mark)
	A1	For the correct value of r
	M1	Uses their value of r to find a value for a
	A1	For a = 200
	A1 cso	For the correct required expression of $\boldsymbol{U_n} = 200\left(\frac{4}{5}\right)^{n-1}$ (must have $\boldsymbol{U_n}$)
(a)ALT	M1	For either correct equation.
	M1	For both correct equations
	dM1	For eliminating r from their equations, reaches a =... (Depends on previous mark)
	A1	For the correct value of a
	M1	Uses their value of a to find a value for r
	A1	For the correct value of r
	A1 cso	For the correct required expression of $U_n = 200\left(\frac{4}{5}\right)^{n-1}$ (must have $\boldsymbol{U_n}$)
(b)	B1	For stating the correct reason, $\lvert r\rvert < 1$ or -1 < r < 1 or "0.8" < 1 or $\left\lvert "\frac{4}{5}" \right\rvert < 1$ **Do not accept** just r < 1 without a correct reason
(c)	M1	For using their value of A /a /first term of their geometric sequence and their r (provided their $\lvert r\rvert < 1$)in a correct formula for the sum to infinity. $S_\infty = \frac{'their\ A'}{1-'(their\ r)'}$
	A1cao	For the correct value of 1000
(d)	M1	Uses their a and r (their r can be $\lvert r\rvert < 1$ or $\lvert r\rvert > 1$) to set the correct formula for the sum of a geometric series > 978 or =978 or ≥ 978 (>978 or =978 or ≥ 978 may be implied by later work.)
	dM1	Attempts to rearrange and achieves $(their\ r)^n$ " < "k (k *is non zero positive constant*) **Allow strict or non-strict inequalities or equation**: <,>, ≥ , ≤ or =
	ddM1	For correctly takes logs (any base)/ln, and achieves a positive value for n, it maybe implied by 17.1…but not 18 **ALT**: trial and error to find a value of n (at least tries n=17 and n=18)
	A1	For n = 18 Must come from **correct** working, achieves n=18 with no incorrect inequalities in their working.

If a candidate does not score the dM1 mark but arrives at an answer of 17.1...
this scores a maximum of M1 as clear use of calculator is not condoned.

Type MS · Sum to infinity of a geometric series

Answer 6 of 13 · Question 8 part c · 2 marks

2024 JUN · Paper 02 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Sum to infinity of a geometric series

Answer 7 of 13 · Question 5 part b · 2 marks

2021 JUN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

5 (a)	$u_2 + u_4 = ar + ar^3 = 212.5$ $u_3 + u_4 = ar^2 + ar^3 = 62.5$	M1
	$\frac{(1+r^2)}{(r+r^2)} = \frac{17}{5}$	M1
	$12r^2 + 17r - 5 = 0$ $(4r-1)(3r+5) = 0$	M1 dM1
	$r = \frac{1}{4} \quad r = -\frac{5}{3}$	A1 (5)
(b)	$r = \frac{1}{4} \Rightarrow a = 800$ So $\frac{a}{1-r} = \frac{800}{\frac{3}{4}} = \frac{3200}{3}$	M1 A1 (2)
		Total 7 marks

Type MS · Sum to infinity of a geometric series

Answer 7 of 13 · Question 5 part b · 2 marks

2021 JUN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Guidance
(a)	M1	For either $ar + ar^3 = 212.5$ or $ar^2 + ar^3 = 62.5$ correct
	M1	For attempting to eliminate a or ar either by division or substitution: e.g. $\frac{ar\left(1+r^2\right)}{ar\left(r+r^2\right)} = \frac{212.5}{62.5} \Rightarrow \frac{\left(1+r^2\right)}{\left(r+r^2\right)} = \frac{212.5}{62.5} = \left(\frac{17}{5}\right)$ An attempt involves some factorisation to eliminate a or ar
	Method 1 – finds a 3TQ	
	M1	For forming a 3TQ in r only using their expressions. $\left(12r^2 + 17r - 5 = 0 \text{ oe}\right)$ Accept for example $150r^2 + 212.5r - 62.5 = 0$
	dM1	For an attempt to solve their 3TQ to give **two** values of r See General Guidance for the definition of an attempt. For example: $(4r-1)(3r+5) = 0 \Rightarrow r = ..., ...$ This mark is dependent on the **FIRST** M mark being awarded
	Method 2 – finds a cubic equation	
	M1	For forming a cubic with a common factor of r in each term. e.g. $12r^3 + 17r^2 - 5r = 0$
	dM1	For factorising their cubic equation to achieve $r\left(12r^2 + 17r - 5\right) = 0$ and for an attempt to solve their 3TQ to give **two** values of r Ignore $r = 0$ if also given. See General Guidance for the definition of an attempt. For example: $(4r-1)(3r+5) = 0 \Rightarrow r = ..., ...$ This mark is dependent on the **FIRST** M mark being awarded
	A1	For the correct values; $r = \frac{1}{4}$ and $r = -\frac{5}{3}$ (reject $r = 0$ if seen earlier)
(b)	M1	Uses their $r = \frac{1}{4}$ [where $\lvert r \rvert < 1$] to find the value of a (800) with the correct formula for the sum of a geometric series to infinity. Condone an incorrect value of a even if they have used $r = \frac{1}{4}$ [The formula is given on page 2 of this booklet]. $S_\infty = \frac{a}{1-r} = \frac{'800'}{1-'\frac{1}{4}'} = ...$
	A1	For the correct value, $S_\infty = \frac{3200}{3}$ or $1066\frac{2}{3}$ Do not accept for example 1066.67 unless the stated value is $1066.\dot{6}$

Type MS · Sum to infinity of a geometric series

Answer 7 of 13 · Question 5 part b · 2 marks

2021 JUN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sum to infinity of a geometric series

Answer 8 of 13 · Question 7 part c · 3 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 4 · find part c in this mark scheme extract

7 (a)	$\frac{ar^6}{ar^3} = r^3 = \frac{e^{\frac{2x+1}{2}}}{e^{x+2}} = \frac{e^x \times e^{\frac{1}{2}}}{e^x \times e^2} = e^{-\frac{3}{2}} \Rightarrow r = e^{-\frac{1}{2}}$ * **ALT** $\frac{e^{x+2}}{r^3} r^6 = e^{\frac{2x+1}{2}} \Rightarrow r^3 = e^{x+\frac{1}{2}-x+2} \Rightarrow r^3 = e^{-\frac{3}{2}} \Rightarrow r = e^{-\frac{1}{2}}$ *	M1M1A1 cso [3] {M1} {M1}{A1} cso [3]
(b)	$ar^3 = ae^{-\frac{3}{2}} = e^{x+2} \Rightarrow a = \frac{e^{x+2}}{e^{-\frac{3}{2}}} = \frac{e^x \times e^2}{e^{-\frac{3}{2}}} = e^{x+\frac{7}{2}}$ oe **ALT** $a = \frac{e^{x+2}}{e^{-\frac{3}{2}}} = e^{x+2+\frac{3}{2}} = e^{x+\frac{7}{2}}$ oe	M1M1A1 [3] {M1}{M1} {A1} [3]

(a)	
M1	Use of $\frac{ar^6}{ar^3} = r^3$
M1	Using $e^{a+b} = e^a \times e^b$ to simplify leading to $r^3 = e^c$ where c is a number
A1 cso	Obtains the **given** answer with no errors in the working
ALT	
M1	For rearranging to make a subject and substituting into the other equation
M1	Using $e^{a-b} = e^a \div e^b$ to simplify leading to $r^3 = e^c$ where c is a number
A1 cso	Obtains the **given** answer with no errors in the working
(b)	
M1	For $a = \frac{e^{x+2}}{e^{-\frac{3}{2}}}$
M1	Using $e^{a+b} = e^a \times e^b$ to simplify to $a = \frac{e^x \times e^2}{e^{-\frac{3}{2}}}$
A1	$e^{x+\frac{7}{2}}$ oe
ALT	
M1	For $a = \frac{e^{x+2}}{e^{-\frac{3}{2}}}$
M1	Using $e^{a+b} = e^a \times e^b$ to simplify to $e^{x+2+\frac{3}{2}}$
A1	$e^{x+\frac{7}{2}}$ oe

Type MS · Sum to infinity of a geometric series

Answer 8 of 13 · Question 7 part c · 3 marks

2020 JAN · Paper 02 (regional R) · extract 2 of 4 · find part c in this mark scheme extract

(c)	$S_\infty = \dfrac{e^{x+\frac{7}{2}}}{1-e^{-\frac{1}{2}}} = \dfrac{e^{x+\frac{7}{2}}}{\frac{e^{\frac{1}{2}}-1}{e^{\frac{1}{2}}}} = \dfrac{e^{x+4}}{e^{\frac{1}{2}}-1} \Rightarrow p = x+4$	M1M1A1 [3]
	ALT 1 $S_\infty = \dfrac{e^{x+\frac{7}{2}}}{1-e^{-\frac{1}{2}}} = \dfrac{e^{x+\frac{7}{2}}}{1-e^{-\frac{1}{2}}} \times \dfrac{e^{\frac{1}{2}}}{e^{\frac{1}{2}}} = \dfrac{e^{x+4}}{e^{\frac{1}{2}}-1} \Rightarrow p = x+4$	{M1}{M1} {A1} [3]
	ALT 2 $S_\infty = \dfrac{e^{x+\frac{7}{2}}}{1-e^{-\frac{1}{2}}} = \dfrac{e^{p}}{e^{\frac{1}{2}}-1} \Rightarrow e^{x+\frac{7}{2}}\left(e^{\frac{1}{2}}-1\right) = e^{p}\left(1-e^{-\frac{1}{2}}\right)$ $\Rightarrow e^{x+4} - e^{x+\frac{7}{2}} = e^{p} - e^{p-\frac{1}{2}} \Rightarrow p = x+4$	{M1}{M1} {A1} [3]

(c)	
M1	Use of $S_\infty = \dfrac{a}{1-r}$
M1	$1-e^{-\frac{1}{2}} = \dfrac{e^{\frac{1}{2}}-1}{e^{\frac{1}{2}}}$
A1	$p = x+4$
ALT 1	
M1	Use of $S_\infty = \dfrac{a}{1-r}$
M1	Multiplying S_∞ by $\dfrac{e^{\frac{1}{2}}}{e^{\frac{1}{2}}}$
A1	$p = x+4$
ALT 2	
M1	Use of $S_\infty = \dfrac{a}{1-r}$
M1	For simplifying to $e^{x+4} - e^{x+\frac{7}{2}} = e^{p} - e^{p-\frac{1}{2}}$ oe
A1	$p = x+4$

Type MS · Sum to infinity of a geometric series

Answer 8 of 13 · Question 7 part c · 3 marks

2020 JAN · Paper 02 (regional R) · extract 3 of 4 · find part c in this mark scheme extract

(d)	$e^{x+\frac{7}{2}} \times \left(e^{-\frac{1}{2}}\right)^{17} > 1.6 \Rightarrow e^x \times e^{\left(\frac{7}{2}-\frac{17}{2}\right)} > 1.6 \Rightarrow e^x > \frac{1.6}{e^{-5}}$	M1
	$\Rightarrow e^x > 237.46105....$	M1
	$\Rightarrow x > \ln(237.46105) \Rightarrow x > 5.4700...$	M1
	$\Rightarrow x = 6$	A1
		[4]
	ALT	
	$e^{x+\frac{7}{2}} \times \left(e^{-\frac{1}{2}}\right)^{17} > 1.6 \Rightarrow e^{x-5} > 1.6$	{M1}
	$\Rightarrow x - 5 > \ln(1.6)$	{M1}
	$\Rightarrow x > \ln(1.6) + 5 \Rightarrow x > 5.4700...$	{M1}
	$\Rightarrow x = 6$	{A1}
		[4]
		Total 13 marks

Type MS · Sum to infinity of a geometric series

Answer 8 of 13 · Question 7 part c · 3 marks

2020 JAN · Paper 02 (regional R) · extract 4 of 4 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sum to infinity of a geometric series

Answer 9 of 13 · Question 3 part b · 2 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

3 (a) (i)	$ar^2 = 5$ or $ar^4 = \frac{5}{2}$ or $5r^2 = \frac{5}{2}$	B1
	$\left(\frac{ar^4}{ar^2}\right) = \frac{\frac{5}{2}}{5}$ or $\frac{5}{r^2} = \frac{\frac{5}{2}}{r^4}$ oe $\rightarrow r$ or $r = \sqrt{\frac{\frac{5}{2}}{5}}$	M1
	$r = \frac{\sqrt{2}}{2}$ oe	A1
(ii)	$a = 10$	A1 [4]
(b)	$S_\infty = \frac{"10"}{1-"\frac{\sqrt{2}}{2}"}$	M1
	$20 + 10\sqrt{2}$	A1 [2]
		Total 6 marks

Type MS · Sum to infinity of a geometric series

Answer 9 of 13 · Question 3 part b · 2 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sum to infinity of a geometric series

Answer 10 of 13 · Question 7 part c · 4 marks

2020 JAN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

7(a)	$\frac{ar^7}{ar^6} = \frac{1152}{192} \quad (= 6) = r$	B1
	4th term $= \frac{192}{6^3}$ or $\frac{1152}{6^4} = \frac{8}{9}$	M1A1 (3)
(b)	$\frac{t_3}{r} + t_3 + rt_3 \Rightarrow \frac{24}{r} + 24 + 24r = -36$	M1A1 NB B1B1 on e-PEN
	$24 + 24r + 24r^2 = -36r$	dM1
	$24r^2 + 60r + 24 = 2r^2 + 5r + 2 = 0$ *	ddM1A1cso (5)
(c)	$2r^2 + 5r + 2 = 0 \Rightarrow (2r + 1)(r + 2) = 0 \Rightarrow r = -\frac{1}{2}$	M1A1
	$S = \frac{a}{1-r} = \frac{24 \div \left(-\frac{1}{2}\right)^2}{1-\left(-\frac{1}{2}\right)}, = 64$	M1,A1 (4)
		[12]

(a)	
B1	Obtain a correct value for r. Fraction need not be simplified
M1	Use their r and either the 7th or 8th term divided by the appropriate power of r to obtain the 4th term as a fraction – no need to simplify
A1	$\frac{8}{9}$
ALT	M1 Find a (=1/243) and use ar^3 A1 Correct answer
(b)	
M1	Use the given information to obtain an equation in r
A1	Correct equation
dM1	Eliminate the fraction
ddM1	Obtain a 3TQ, terms in any order
A1cso	Reach the given result with no errors in the working
(c)	
M1	Solve the given quadratic by any valid method. Must reach a value of r
A1	Correct value of r (Ignore second answer if given)
M1	Use the formula for the sum to infinity with their r **provided** $\lvert r \rvert < 1$. a must be 24 divided by (their r)^2
A1	Correct answer.

Type MS · Sum to infinity of a geometric series

Answer 10 of 13 · Question 7 part c · 4 marks

2020 JAN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question Number	Scheme	Marks
	$\frac{dy}{}$	

Type MS · Sum to infinity of a geometric series

Answer 11 of 13 · Question 7 part b · 2 marks

2023 NOV · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

7(a)	$\left(\frac{ar^2}{a} = r^2 =\right) \frac{\frac{2704}{625}}{16}$ oe	M1
	$(r =) \pm \frac{13}{25}$ oe	A1 [2]
(b)	$(S_\infty =) \frac{16}{1 - "\frac{13}{25}"} = \frac{100}{3}$ oe	M1A1 [2]
(c)	$\frac{16\left(1 - \left("\frac{13}{25}"\right)^n\right)}{1 - "\frac{13}{25}"} > 33 \left(\Rightarrow 1 - \left(\frac{13}{25}\right)^n > 0.99\right)$ oe	M1
	$\Rightarrow \left("\frac{13}{25}"\right)^n < 0.01$ oe	dM1
	$n \lg\left("\frac{13}{25}"\right) < \lg(0.01)$ oe	M1
	$\Rightarrow n > \frac{\lg(0.01)}{\lg\left("\frac{13}{25}"\right)} \Rightarrow n > 7.04...\text{oe} \Rightarrow (n =) 8$	dddM1A1 [5]
		Total 9 marks

Type MS · Sum to infinity of a geometric series

Answer 11 of 13 · Question 7 part b · 2 marks

2023 NOV · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For any correct unsimplified expression for r or r^2.
	A1	For $r = \pm\frac{13}{25}$ oe
(b)	M1	Uses the correct formula for the sum to infinity of a geometric series using their positive $r < 1$ and the given first term.
	A1	For $\frac{100}{3}$ oe . Allow 33.33……, 33.3^r, $33.\dot{3}$ etc. The question demand is for an exact answer so there must be some indication of recurrence. This could be min 1 dp with dots after ie 33.3….., but cannot be a terminating decimal.
(c)	M1	Uses the correct formula for the sum of a geometric series to set up an **inequality or equation** in terms of n, *ft* their **positive** $r < 1$, must be using $a = 16$ Allow < or > or =
	dM1	For simplifying (allow errors in simplification) their **inequality or equation** in n to the form $\left("\frac{13}{25}"\right)^n < d \quad d \neq 0$. Allow $"\frac{13}{25}"$ to be their positive $r < 1$. Allow < or > or =. Dependent on the 1st method mark.
	M1	Takes logarithms (any base) of their **exponential equation or inequality <u>correctly</u>** on both sides and correctly uses the power law to reach $n\lg(a) < \lg(b)$. or 'de-logs' their **exponential equation or inequality <u>correctly</u>** to get $\log_e d < e$ This isn't a dependent mark, but the candidate must be able to correctly take logs of both sides and use the power law or correctly de-log their equation/inequality. r can be any value carried through from part (a) for this mark. Allow < or > or =, the inequality doesn't need to have been reversed at this point.
	dddM1	For finding a value for n setting up the correct inequality (with their positive $r < 1$) from the beginning and where the inequality sign has been reversed at the appropriate point during their work. Dependent on all previous method marks. If candidates give a final answer of (n =) 8 – this mark can be implied even if the inequality sign is not correctly reversed.
	A1	For (n =) 8

Type MS · Sum to infinity of a geometric series

Answer 11 of 13 · Question 7 part b · 2 marks

2023 NOV · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Sum to infinity of a geometric series

Answer 12 of 13 · Question 2 part whole · 6 marks

2023 JUN · Paper 02 · extract 1 of 4 · find part whole in this mark scheme extract

2	$n=1 \Rightarrow a=8^{(1-2)}=8^{-1}$ oe eg $\frac{1}{8}$	B1
	$n=2 \Rightarrow ar=8^{(1-4)}=8^{-3} \Rightarrow r=\frac{8^{-3}}{8^{-1}}=8^{-2}$ oe eg $\frac{1}{64}$	M1A1
	$(S_\infty =)\frac{8^{-1}}{1-8^{-2}}=\frac{2^{-3}}{\frac{63}{2^6}}=\frac{8}{63}$ oe eg $\frac{\frac{1}{8}}{1-\frac{1}{64}}\left(=\frac{\frac{1}{8}}{\frac{63}{64}}=\frac{1}{8}\times\frac{64}{63}\right)$	M1dM1
	$\frac{8}{63}$ oe or $p=8,\ q=63$ oe	A1
		Total 6 marks

Type MS · Sum to infinity of a geometric series

Answer 12 of 13 · Question 2 part whole · 6 marks

2023 JUN · Paper 02 · extract 2 of 4 · find part whole in this mark scheme extract

Mark	Notes
M1	For multiplying both numerator & denominator of $\frac{(a+2\sqrt{5})}{(3-\sqrt{5})}$ through by $\frac{(3+\sqrt{5})}{(3+\sqrt{5})}$ to give $\frac{3a+a\sqrt{5}+6\sqrt{5}+10}{9-5}$. Allow one error on the numerator. The denominator must be correct.
M1	For correctly equating **their** coefficients with $\frac{11+b\sqrt{5}}{2}$. **Although this is not a dependent mark, there must be at least one equation in *a* and *b***
M1	For a complete and correct attempt to solve one of their equations to find a value for a **or** a value for b. **Although this is not a dependent mark, there must be at least one equation in *a* and *b*.**
A1	For either $a = 4$ or $b = 5$
A1	For both $a = 4$ and $b = 5$
Students may also multiply by $-3-\sqrt{5}$ This produces the correct answer and is the same as the main MS, but all the terms are negative. Mark to exactly the same principles.	
ALT	
M1	Correctly removes the denominators from the given equation and multiplies out as shown to give the equation $2a+4\sqrt{5}=33+3b\sqrt{5}-11\sqrt{5}-5b$ Allow one error.
M1	For correctly equating **their** coefficients on either side of the equation. **Although this is not a dependent mark, there must be at least one equation in *a* and *b***
M1	For a correct and complete attempt to solve one of their equations to find a value for a **or** a value for b. **Although this is not a dependent mark, there must be at least one equation in *a* and *b***
A1	For either $a = 4$ or $b = 5$
A1	For both $a = 4$ and $b = 5$

Type MS · Sum to infinity of a geometric series

Answer 12 of 13 · Question 2 part whole · 6 marks

2023 JUN · Paper 02 · extract 3 of 4 · find part whole in this mark scheme extract

Question	Scheme	Marks
2	$n=1 \Rightarrow a=8^{(1-2)}=8^{-1}$ oe eg $\frac{1}{8}$	B1
	$n=2 \Rightarrow ar=8^{(1-4)}=8^{-3} \Rightarrow r=\frac{8^{-3}}{8^{-1}}=8^{-2}$ oe eg $\frac{1}{64}$	M1A1
	$(S_\infty =)\frac{8^{-1}}{1-8^{-2}}=\frac{2^{-3}}{\frac{63}{2^6}}=\frac{8}{63}$ oe eg $\frac{\frac{1}{8}}{1-\frac{1}{64}}\left(=\frac{\frac{1}{8}}{\frac{63}{64}}=\frac{1}{8}\times\frac{64}{63}\right)$	M1dM1
	$\frac{8}{63}$ oe or $p=8,\ q=63$ oe	A1
		Total 6 marks

Mark	Notes
B1	For $a=8^{-1}$ oe
M1	For substituting $n = 2$ into the expression for nth term to find a value for ar **and** dividing by a to find r. This mark can be implied by a correct value for r.
A1	For $r=8^{-2}$ oe
M1	For applying the correct formula for the sum to infinity of a convergent geometric series for their values of a and r, providing $\lvert r\rvert<1$ They must be using values they've attained or stated for a and r.
dM1	For a correct attempt to use an index law with their expression to obtain the required form or a correct attempt to divide their fractions. Dependent on previous method mark.
A1	For the correct answer in the required form any equivalent with p and q integers is acceptable.
In this question, the final dM1 may implied from a correct substitution of their values of r and a, evaluated correctly, if working isn't shown. You may have to check their final answer. Eg $r=\frac{1}{4}, a=\frac{1}{8}$ $(S_\infty =)\frac{\frac{1}{8}}{1-\frac{1}{4}}=\frac{1}{6}$ is M1 dM1 A0 because the $\frac{1}{6}$ is correct for their a and their r	

Type MS · Sum to infinity of a geometric series

Answer 12 of 13 · Question 2 part whole · 6 marks

2023 JUN · Paper 02 · extract 4 of 4 · find part whole in this mark scheme extract

Question	Scheme	Marks

Type MS · Sum to infinity of a geometric series

Answer 13 of 13 · Question 4 part ii · 7 marks

2025 JUN · Paper 02 (regional R) · extract 1 of 3 · find part ii in this mark scheme extract

Number		
4 (i)	$a=3$ or $r=2$	B1
	$\left(\sum_{k=1}^{17}3(2)^{k-1}\right)=\frac{3\left(1-(2)^{17}\right)}{1-2}=(393213)$ $\left(\sum_{k=1}^{5}3(2)^{k-1}\right)=\frac{3\left(1-(2)^{5}\right)}{1-2}=(93)$	M1
	"393213"–"93"	dM1
	393120	A1
		[4]
ALT	$a=3\times2^5\,(=96), r=2$	B1
	$\left(\sum_{k=6}^{17}3(2)^{k-1}\right)=\frac{\left(3\times2^5\right)\left(1-(2)^{\text{"11 or 12"}}\right)}{1-2}$	M1
	$\frac{\left(3\times2^5\right)\left(1-(2)^{12}\right)}{1-2}$	M1
	393120	A1
		[4]
(ii)	$ar^2=\frac{7}{13}$ or $ar^8=\frac{448}{9477}$	M1
	$\frac{ar^8}{ar^2}=\frac{\frac{448}{9477}}{\frac{7}{13}}\left(=\frac{64}{729}\right)\Rightarrow r^6=$	dM1
	$r=\sqrt[6]{\left(\frac{\frac{448}{9477}}{\frac{7}{13}}\right)}$	M1
	$r=\frac{2}{3}$ $a=\frac{63}{52}$	A1 A1
	$S_\infty=\frac{\text{"}\frac{63}{52}\text{"}}{1-\text{"}\frac{2}{3}\text{"}}$	M1
	$=\frac{189}{52}$ oe	A1
		[7]
		Total 11 marks

Type MS · Sum to infinity of a geometric series

Answer 13 of 13 · Question 4 part ii · 7 marks

2025 JUN · Paper 02 (regional R) · extract 2 of 3 · find part ii in this mark scheme extract

Part	Mark	Additional Guidance
(i)	B1	For either value of a or r correct, stated or implicit in working.
	M1	For correctly substituting into the formula for either sum. Allow use of their a and their r with any numbers it is clear they intend to be a and r
	dM1	For their 2 sums subtracted, the correct way round. Dependent on the previous method mark. **They must be using $a = 3$ for this mark.**
	A1	For 393120
ALT	B1	For either value of a or r correct, stated or implicit in working.
	M2	For correctly substituting into the formula for the sum to n terms. **Must be using $n = 12$ and $a = 96$, may use their r.** For M1 students may use $n = 11$ or 12 and allow use of their a and their r with any numbers it is clear they intend to be a and r
	A1	For 393120
	Students who complete this by adding every term B1 for the first term correct M2 for all terms written out and added (we are not expected to evaluate them), there is no M1 mark A1 correct answer and as usual, can imply the M and B marks. **A fully correct answer with little or no working will score 4 marks**	
(ii)	M1	For either correct equation.
	dM1	For correctly using their equations to eliminate a (or r) We will condone $\frac{ar^9}{ar^3}\left(=\frac{\frac{448}{9477}}{\frac{7}{13}}=\frac{64}{729}\right)\Rightarrow r^6=$ For this mark if the candidate works correctly to find r
	M1	Rearranges their equation, allowing one processing error, to find a value for r (or a). Do not award the mark if a student moves from r^6 to r without showing what they are doing if r is incorrect. We will condone correct r from $\frac{ar^9}{ar^3}\left(=\frac{\frac{448}{9477}}{\frac{7}{13}}=\frac{64}{729}\right)\Rightarrow r=$ if the candidate works correctly to find r
	A1	Either r or a correct.
	A1	Both r and a correct.
	M1	Correct substitution of their values for a and r into a correct formula for S_∞
	A1	$\frac{189}{52}$ oe

Type MS · Sum to infinity of a geometric series

Answer 13 of 13 · Question 4 part ii · 7 marks

2025 JUN · Paper 02 (regional R) · extract 3 of 3 · find part ii in this mark scheme extract

Question number	Scheme	Marks